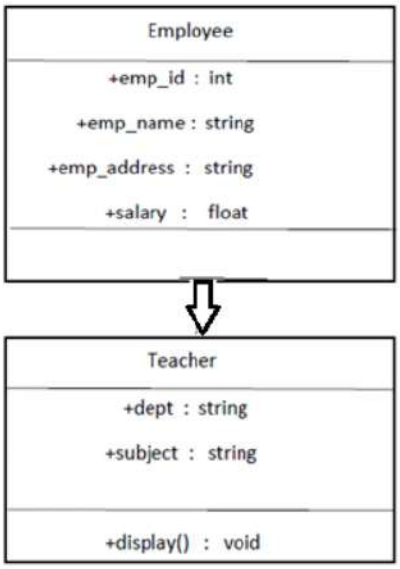
PROGRAM NO – 34

AIM: To develop a Java program to create a class 'employee' with data members empid, name, salary,
address and constructors to initialize the data members. Create another class 'teacher' that inherit the properties of class employee and contain its own data members department, subjects taught and constructors to initialize these data members and also include display function to display all the data members. Use array of objects to display details of N teachers.



PROGRAM

```
import java.util.Scanner;
class Employee                          //Parent class
{
   int empid;
   String name,address,subject;
   double salary;

   Employee(int no, String na, double sal, String addr)  // Parent class constructor
   {
      empid = no;
      name = na;
      salary = sal;
      address = addr;
   }
}
```

```
class Teacher extends Employee                    //Child class
{
  String dept,sub;
  Teacher(int no, String na, double sal, String addr, String dep,String sub) //Child class
constructor
  {
    super(no,na,sal,addr);    // Passing data to parent class constructor
    dept= dep;
    subject= sub;
  }
  void display()
  {
    System.out.println("Employee id: "+empid);
    System.out.println("Name: "+name);
    System.out.println("Salary: "+salary);
    System.out.println("Address: "+address);
    System.out.println("Department: "+dept);
    System.out.println("Subject: "+subject);
  }
}
class EmpInheritance
{
   public static void main(String[] args)
  {
    System.out.println("\nEnter the no. of Employees");
    Scanner s = new Scanner(System.in);
    int num = s.nextInt();
    Teacher arr[]=new Teacher[num];
    for(int i =0;i<num;i++)
    {
```

```
            System.out.println("\nEnter Employee-"+(i+1)+" details: id,Employee Name, Salary,
                                Address, Department,Subject");
        int empid=s.nextInt();
        String name=s.next();
        double salary=s.nextDouble();
        String address=s.next();
        String dept=s.next();
        String subject=s.next();
        arr[i]=new Teacher(empid,name,salary,address,dept,subject); // Passes values to child class
constructor
    }
    System.out.println("\n**Informations of all the employee's**");
    for(int i=0;i<num;i++)
    {
        System.out.println("\nEmployee-"+(i+1)+" details");
        System.out.println("----------");
        arr[i].display();
    }
    s.close();
  }
}
```

Leave 1 page for output

NOTES

super keyword

- When a superclass has its fields declared as private, these fields are private to the superclass and are not accessible by its subclasses using the conventional style.
- Whenever a subclass needs to refer to its immediate superclass, it can do so by using the keyword **super**. We use super in two ways:

• To call superclass constructor: A subclass can call a constructor defined by its superclass as
follows:

```
super(arg-list );
```

- Here, arg-list specifies arguments, if any, needed by the constructor of its superclass.
- super() must be the 1st statement inside a subclass's constructor.
- Eg: consider the program below.

```
class Animal
{
        Animal ( )
        {
                System . out . p r i n t l n ( " Animal  i s  c r e a t ed " ) ;
        }
}
class Dog extends Animal
{
        Dog ( )
        {
                super ( ) ;
                System . out . p r i n t l n ( "dog  i s  c r e a t ed " ) ;
        }
}
class TestSuper1
{
        public static void main ( String args [ ] )
        {
                Dog d=new Dog ( ) ;

        }
}
```

Output

animal is created

dog is created

PROGRAM NO - 35

AIM: To develop a Java program to demonstrate multilevel Inheritance. Create a class "Principal"(name) which is inherited by the class "Teacher" (tchrId) which in turn is inherited by class "Student" (rno,display())

PROGRAM

```
class Principal
{
    String name="Rakesh";
}
class Teacher extends Principal
{
    int tchrId=56;
}
class Student extends Teacher
{
    int rno=78;
    void display()
    {
        System.out.println("Principal name :"+name);
        System.out.println("Teacher-id :"+tchrId);
        System.out.println("Student-roll no :"+rno);
    }
}
class Multilevel
{
    public static void main(String args[])
    {
        Student ob=new Student();
        ob.display();
    }
}
```

NOTES

Multilevel Inheritance

When there is a chain of inheritance, it is known as multilevel inheritance.
"We create objects for the class at the lowest level"
In the example given below, Animal class is inherited by Dog class and Dog class is inherited by BabyDog class.

```
class Animal
{
        void eat( )
        {
                System.out.p r i n t l n ( " e a t ing . . . " ) ;
        }
}
class Dog extends Animal
{
        void bark( )
        {
                System.out.p r i n t l n ( " barking . . . " ) ;
        }
```

```
}
class BabyDog extends Dog
{
        void weep( )
        {
                System.out.p r i n t l n ( "weeping . . . " ) ;
        }
}
class TestInheritance
{
        public static void main ( St ring args [ ] )
        {
                BabyDog d = new BabyDog ( ) ;
                d.weep ( ) ;
                d.bark ( ) ;
                d.eat( ) ;
        }
}
```

PROGRAM NO - 36

AIM: To develop a Java program to demonstrate multilevel Inheritance. Create a class "Manager"(salary) which is inherited by the class "TeamLead" (teamName) which in turn is inherited by class "Developer" (prjcId,putdata())

PROGRAM

Leave space as was needed for Pgm-35

PROGRAM NO – 37

AIM: To develop a Java program to demonstrate use of "super" keyword to access super class constructor. Create a class "SuperParent" (rollno) which is inherited by the class "SuperChild"(name,display()).

PROGRAM

```
class SuperParent
{
    int rollno;
    SuperParent(int rno)
    {
        rollno=rno;
    }
}
class SuperChild extends SuperParent
{
    String name;
    SuperChild(int rno, String name1)
    {
        super(rno); //Calling super class constructor
        name=name1;
    }
    void display()
    {
        System.out.println("Roll no.:"+rollno);
        System.out.println("Name.:"+name);
    }
}
class SuperDemo
{
    public static void main(String args[])
    {
        SuperChild ob=new SuperChild(23,"Deepthy");
        ob.display();
    }
}
```

OUTPUT

PROGRAM NO – 38

AIM: To develop a Java program to demonstrate use of "super" keyword to access super
class constructor. Create a class "Parent" (pid) which is inherited by the class "Child"(course,putdata()).

PROGRAM

Leave space as was needed for Pgm-37

PROGRAM NO – 39

AIM: To develop a Java program to create a class "Person" with data members name, gender, address, age & a constructor to initialize data member & another class 'Employee' that inherits the properties of class Person & also contains its own data members like empid, company, name, qualification, salary & its own constructor. Create another class "Teacher" that inherits properties of class Employee & contain its own data members like subject, department, teacher id & also contain constructors & methods to display the data members. Use array of objects to display details of the teachers. (Multilevel Inheritance)

PROGRAM

```
import java.util.Scanner;
class Person
{
    String pname, address, gender;
    int age;
    Person(String pname1, String address1, String gender1, int age1)
    {
        pname=pname1;
        address=address1;
        gender=gender1;
        age=age1;
    }
}
class Employee extends Person
{
    String emp_id, cp_name, qualification;
    float salary;
    Employee(String pname, String address, String gender, int age,
              String emp_id, String cp_name,String qualification,
              float salary)
    {
        super(pname, address, gender, age);
        this.emp_id = emp_id;
        this.cp_name = cp_name;
        this.qualification = qualification;
        this.salary = salary;
    }
}
class Teacher extends Employee
{
    int teach_id;
    String department, subject;
    Teacher(int teach_id, String department, String subject, String
            emp_id, String cp_name,String qualification, float salary,
            String pname, String address, String gender, int age)
    {
        super(pname, address, gender, age, emp_id, cp_name,
              qualification, salary);
        this.teach_id = teach_id;
        this.department = department;
        this.subject = subject;
    }
```

```java
public void display()
 {
       System.out.println("Name: " + pname);
        System.out.println("Gender: " + gender);
        System.out.println("Address: " + address);
        System.out.println("Age: " + age);
        System.out.println("Employee ID: " + emp_id);
        System.out.println("Company Name: " + cp_name);
        System.out.println("Salary: " + salary);
        System.out.println("Qualification: " + qualification);
        System.out.println("Teacher ID: " + teach_id);
        System.out.println("Department: " + department);
        System.out.println("Subject Taught: " + subject);
    }
}
    class MultilevelDemo
    {
      public static void main(String[] args)
      {
        Scanner s = new Scanner(System.in);
        System.out.print("Enter number of employees:");
        int n = s.nextInt();
        s.nextLine();
        Teacher empobj[] = new Teacher[n];
        for (int i = 0; i < n; i++)
        {
            System.out.println("Enter Employee-"+(i+1)+" details");
            System.out.println("................................");
            System.out.print("Name:");
            String pname = s.nextLine();
            System.out.print("Gender:");
            String gender = s.nextLine();
            System.out.print("Address:");
            String address = s.nextLine();
            System.out.print("Age:");
            int age = s.nextInt();
            s.nextLine();
            System.out.print("Employee ID:");
            String emp_id = s.nextLine();
            System.out.print("Company name:");
            String cp_name = s.nextLine();
            System.out.print("Salary:");
            float salary = s.nextFloat();
            s.nextLine();
            System.out.print("Qualification:");
            String qualification = s.nextLine();
            System.out.print("Teacher ID:");
            int teach_id = s.nextInt();
            s.nextLine();
            System.out.print("Department:");
            String department = s.nextLine();
            System.out.print("Subject taught:");
            String subject = s.nextLine();
```

```
            empobj[i] = new Teacher(teach_id, department, subject,
emp_id,
                    cp_name, qualification, salary,pname, address,
gender, age);
        }
        System.out.println("Employee details");
        System.out.println("-----------------");
        for (int i = 0; i < n; i++)
        {
            System.out.println("Employee-"+(i+1));
            System.out.println("--------------");
            empobj[i].display();
        }
      }
    }
```

Leave 1 page for output

NOTES

"this" **Keyword in Java**

In Java, the `this` keyword refers to the **current instance of a class**. It is mainly used to differentiate between instance variables and parameters, call constructors, and refer to the current object.

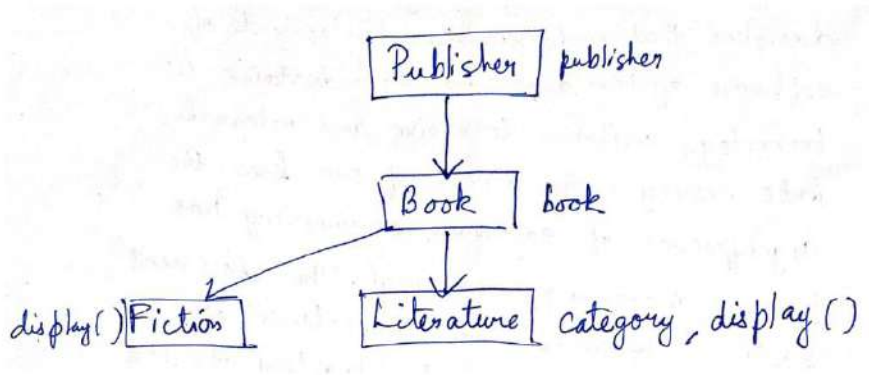
To refer to instance variables when they are shadowed by parameters

```
class Employee
{
  String name;
  int age;
  Employee(String name, int age)
  {
    this.name = name; // Refers to the instance variable
    this.age = age;
  }
  void display()
  {
    System.out.println("Name: " + this.name + ", Age: " + this.age);
  }
}
public class Main
{
  public static void main(String[] args)
  {
```

```
        Employee e = new Employee("John", 25);
        e.display();
    }
}
```

PROGRAM NO – 40

AIM: To develop a Java program to create classes : publisher inherited by book which is inherited by literature and fiction classes. Read the information and print the details of books from either of the category using inheritance.



PROGRAM

```
import java.util.Scanner;
class Publisher
{
    String pubName;
    Publisher(String pub)
    {
        pubName=pub;
    }
}
class Book extends Publisher
{
    String book;
    Book(String pub,String bk)
    {
        super(pub);
        book=bk;
    }
}
class Literature extends Book
{
    String category;
    Literature(String pub, String bk, String ctgry)
    {
        super(pub, bk);
        category=ctgry;
    }
    void display()
    {
        System.out.println("Publisher :"+pubName);
        System.out.println("Book :"+book);
        System.out.println("Category: "+category);
    }
}
class Fiction extends Book
{
    String category;
    Fiction(String pub, String bk, String ctgry)
    {
        super(pub, bk);
        category=ctgry;
    }
    void display()
    {
        System.out.println("Publisher :"+pubName);
```

```
        System.out.println("Book :"+book);
        System.out.println("Category: "+category);
    }
}
public class BookDetails
{
    public static void main(String[] args)
    {
        Scanner s= new Scanner(System.in);
        System.out.println("Enter Literature Book Details");
        System.out.println("Book name");
        String b =s.next();
        System.out.println("Publisher ");
        String pub =s.next();
        System.out.println("Category ");
        String c =s.next();
        Literature ob1=new Literature(b,pub,c);

        System.out.println("Enter Fiction Book Details");
        System.out.println("Book name");
        String b1 =s.next();
        System.out.println("Publisher ");
        String p1 =s.next();
        System.out.println("Category ");
        String c1 =s.next();
        Fiction ob2=new Fiction(b1,p1,c1);

        System.out.println("\n** Literature Book**");
        System.out.println("----------------------");
        ob1.display();

        System.out.println("\n**Fiction Book**");
        System.out.println("----------------------");
        ob2.display();

        s.close();
    }
}
```

PROGRAM NO – 41

AIM: To develop a Java program to create a class "Manager" (manager name) which is inherited by

"Employee" (employee name, display()) and "Staff" (staff name, display()) .
(Hierarchical Inheritance)

PROGRAM

```
import java.util.Scanner;
class Manager
{
    String mngrName;
    Manager(String mname)
    {
        mngrName=mname;
    }
}
class Employee extends Manager
{
    String empName;
    Employee(String mname,String ename)
    {
        super(mname);
        empName=ename;
    }
    void display()
    {
        System.out.println("Manager :"+mngrName);
        System.out.println("Employee :"+empName);

    }
}
class Staff extends Manager
{
    String stfName;
    Staff(String mname, String sname)
    {
        super(mname);
        stfName=sname;
    }
    void display()
    {
        System.out.println("Manager :"+mngrName);
        System.out.println("Staff :"+stfName);

    }
}
public class HierarchicalDemo
{
    public static void main(String[] args)
    {
        Scanner s= new Scanner(System.in);
        System.out.println("Enter Details");
        System.out.print("Manager name :");
        String m =s.next();
        System.out.print("Employee Name :");
        String e =s.next();
        System.out.print("Staff name : ");
        String st =s.next();

        Employee ob1=new Employee(m,e);
        System.out.println("Manager-Employee Details");
        System.out.println("-------------------------");
        ob1.display();
```

```
        Staff ob2=new Staff(m,st);
        System.out.println("Manager-Staff Details");
        System.out.println("-------------------------");
        ob2.display();

        s.close();
    }
}
```

PROGRAM NO – 42

AIM: To develop a Java program to create a class "Hospital" (hospital name) which is inherited by "Doctor" (doctor name, putdata()) and "Nurse" (nurse name, putdata()) . (Hierarchical Inheritance)

PROGRAM

Leave space as was needed for Pgm-41

PROGRAM NO – 43

AIM: To develop a Java program to create a class "Sport" which is inherited by "Student" class. Create another class "Result" which is inherited from "Student". Display the academic and sports score of a student.

PROGRAM

```
import java.util.Scanner;
class Sports
{
    String sport;
    int rating;
    Sports(String sp, int rt)
    {
        sport=sp;
        rating=rt;
    }
}
class Student extends Sports
{
    String grade;
    double overall_per;
    Student(String sp, int rt,String gd, double per )
    {
        super(sp,rt);
        grade = gd;
        overall_per = per;
    }
}
class Result extends Student
{
    Result(String sp, int rt,String gd, double per )
    {
        super(sp,rt,gd,per);
    }
    void display()
    {
        System.out.println("\nSports Details of Student");
        System.out.println("----------------------------");
        System.out.println("Sport :"+sport);
        System.out.println("Rating :"+rating);
        System.out.println("\nAcademic Details of Student");
        System.out.println("----------------------------");
        System.out.println("Academic Grade :"+grade);
        System.out.println("Overall percentage :"+overall_per);
    }
}
class School
{
    public static void main(String[] args)
    {
        Scanner sc =new Scanner(System.in);
        System.out.println("Enter the Sports Details of Student");
        System.out.print("Sport: ");
        String s=sc.next();
        System.out.print("Sport Rating out of 10: ");
        int r=sc.nextInt();
        System.out.println("Enter the Academic Details of Student");
        System.out.print("Academic Grade(A-D): ");
        String g=sc.next();
        System.out.print("Overall percentage: ");
        double p=sc.nextDouble();
```

```
        sc.close();
        Result obj= new Result(s,r,g,p);
        obj.display();
    }
}
```

Leave 1 page for output

PROGRAM NO – 44

AIM: To develop a Java program to create an interface "Vehicle"(String name,msg()) which is implemented by a class "Car" (int car no.).

PROGRAM

```
interface Vehicle
{
    String NAME="i10";
    void msg();
}
class Car implements Vehicle
{
    int cno=6789;
    public void msg()
    {
        System.out.println("Vehicle name: "+NAME);
        System.out.println("Vehicle no.: "+cno);
    }
}
class InterfaceDemo
{
    public static void main(String args[])
    {
        Car ob=new Car();
        ob.msg();
    }
}
```

NOTES

- **Abstract class**: is a restricted class that cannot be used to create objects (to access it, it must be inherited from another class).
- **Abstract method**: It can only be used in an abstract class, and it does not have a body.

Interfaces

- An interface is completely an "abstract class" that is used to group related methods with empty bodies.
- An interface in Java is a blueprint of a class.
- It can have static constants and abstract methods.
- Interfaces are also used to implement multiple inheritance in Java.
- Java classes (including abstract classes) does not support multiple inheritance.

Main features of an interface

- It cannot have a method body.
- It can have static constants

- Java Interface also represents the IS-A relationship.
- It cannot be instantiated just like the abstract class.
- We can have default and static methods in an interface.
- We can have private methods in an interface.

Declaration of an interface

- An interface is declared using the interface keyword.
- It provides total abstraction; means all the methods in an interface are declared with the empty body,
  and all the fields are public, static and final by default.
- A class that implements an interface must implement all the methods declared in the interface.

Syntax

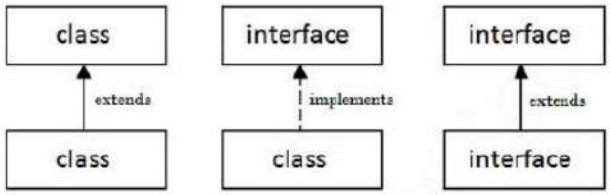
```
interface < Interface_name>
{
        //declare constant variables;
        //declare abstract methods
}
```

Class implementing an interface

```
class <Classname> implements < Interface_name>
{
        …..
        //Definition of abstract methods of interface
}
```

Relationship between classes and interfaces

A class extends another class, an interface extends another interface, but a class implements an interface.



PROGRAM NO – 45

AIM: To develop a Java program to create an interface "Shape"(int length,display()) which is implemented by a class "Rectangle" (int breadth).

PROGRAM

Leave space as was needed for Pgm-44